

Differential Analysis of LLMs in Text2SQL Generation

Jaykumar Patel
patel.jay4802@utexas.edu

An-Jung Huang
ajhuang@utexas.edu

Ryan Kolm
ryankolm@utexas.edu

1 Project Proposal

As large language models (LLMs) continue to evolve and improve, an interesting application of LLMs is in the generation of SQL queries from natural language. This is known as Text2SQL. In our project, we will compare the performance of different LLMs in generating SQL queries from natural language. We will explore 2 different frameworks: **Agentless** and **Agentic**.

1. **Agentless** frameworks are those that use an LLM to generate the response directly. This is synonymous with the use of a chatbot. For this framework, we will provide the schemas of every table in the database and the natural language query to the LLM. Then we will run the generated SQL query and evaluate the results.
2. **Agentic** frameworks are those where there are distinct agents that each play a different role in the generation of the response. For this framework, we will have 2 agents: a **Schema Agent** and a **Query Agent**. The **Schema Agent** will be responsible for retrieving the schemas of the relevant tables from the database given the natural language query. The **Query Agent** will be responsible for generating the SQL query given the relevant schemas and the natural language query. Then we will run the generated SQL query and evaluate the results.

The SQL database that we plan to use is the Chinook sample database, which is a sample database provided by SQLite. For each of the tables in the database, we will create the Schema, which will be used as the context passed to the LLMs. Additionally, we will determine 20 natural language queries, their corresponding SQL queries, and the results of executing the SQL queries. Using that benchmark dataset as ground truth, we will perform a differential analysis of the performance of 3 LLMs for the 2 frameworks. We will analyze the ability of the LLMs to generate correct SQL queries, as well as the differences in performance between the LLMs and frameworks. The LLMs we plan to use are: ChatGPT, Gemini, and Llama.

Note: The SQL database, agentic framework, and LLMs are subject to change.